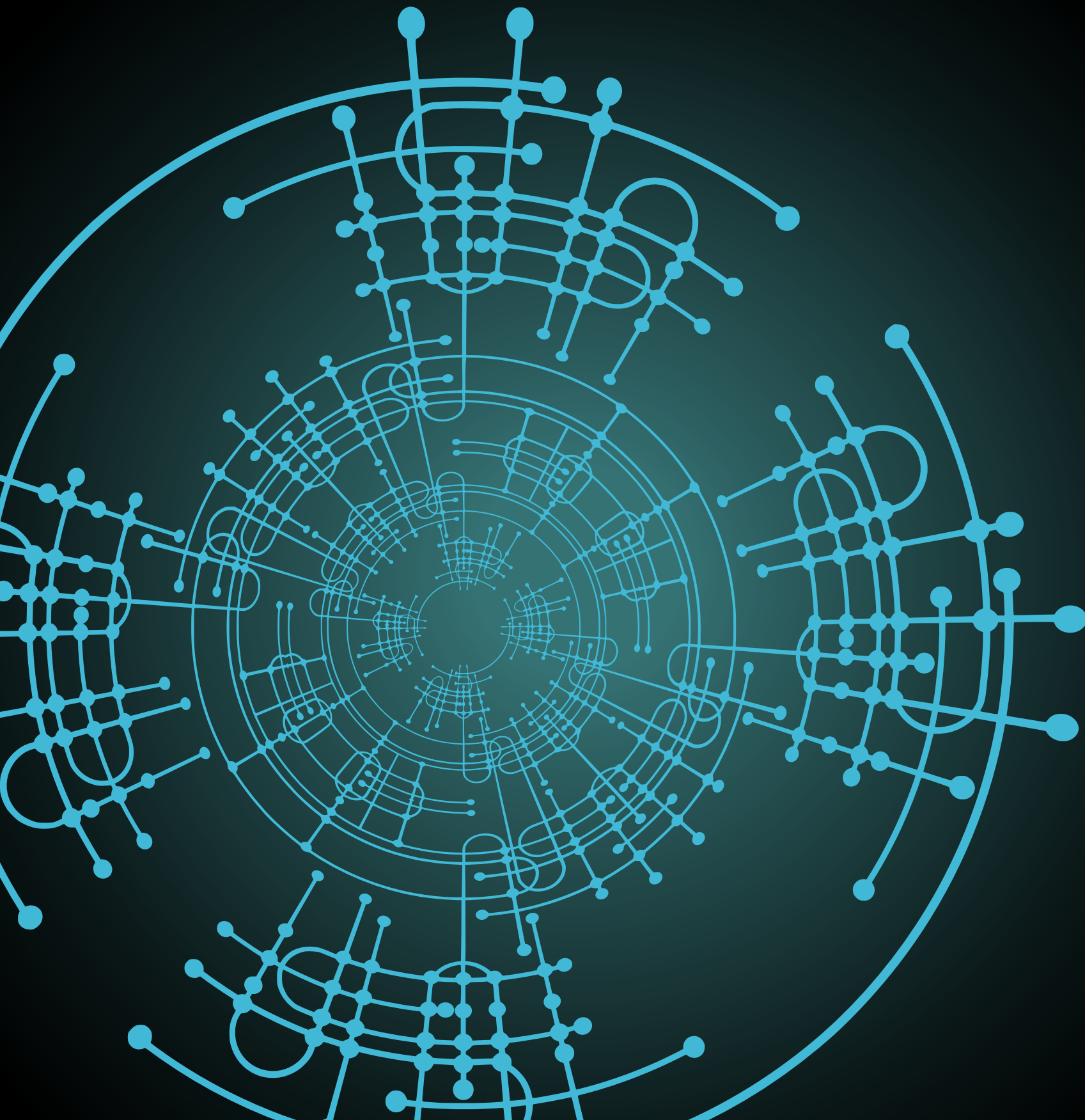


Project Report : Fundamentals of AI & ML



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Course Code : CSA2001

Topic : Smart Expense Tracker

(Machine learning Project)



INTRODUCTION

Managing personal finances is a common challenge, especially for students who have limited budgets and irregular spending habits. Many individuals fail to track their daily expenses, leading to poor financial decisions and lack of savings.

This project, Smart Expense Tracker, is developed to solve this real-world problem by providing an easy-to-use platform where users can record, analyze, and predict their expenses. The application integrates basic Artificial Intelligence and Machine Learning concepts to provide insights and assist users in better financial planning.

PROBLEM STATEMENT

Students and individuals often:

- Do not track their expenses regularly
- Lack awareness of spending pattern
- Fail to plan savings effectively

This leads to:

- Overspending
- Poor budgeting
- Financial stress

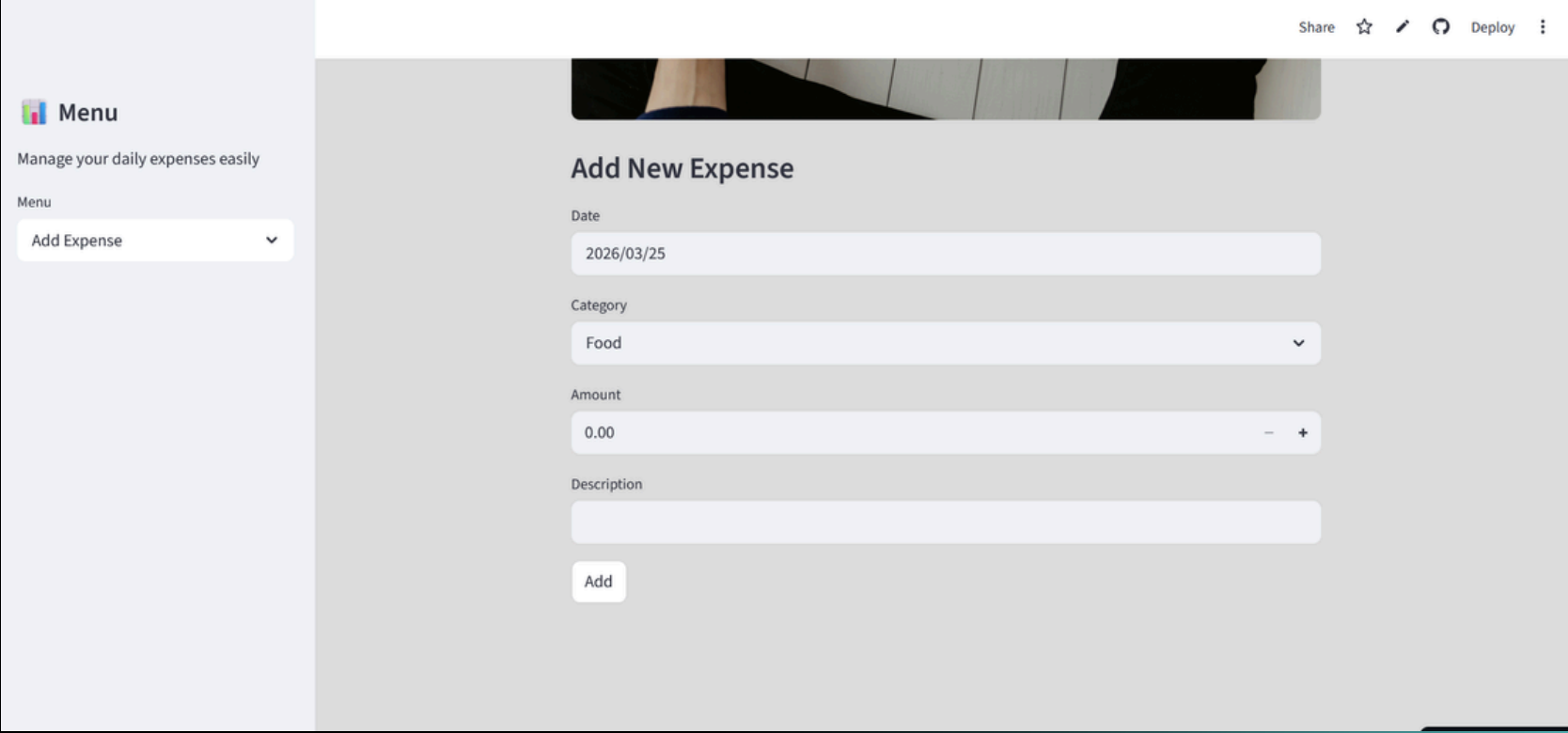
Features of the Project

Add Expense

The application allows users to easily record their daily expenses by entering details such as date, category, amount, and description. This feature ensures proper tracking of all financial activities in a structured manner.

It helps users maintain a habit of recording expenses regularly, which is the first step toward effective financial management.

Interface



The screenshot displays the 'Add New Expense' form within a web application. The interface is clean and modern, with a light gray background. On the left, there is a sidebar menu with the title 'Menu' and the subtitle 'Manage your daily expenses easily'. Below this, there is a dropdown menu labeled 'Menu' with the option 'Add Expense' selected. The main content area is titled 'Add New Expense' and contains several input fields: 'Date' (with the value '2026/03/25'), 'Category' (with the value 'Food'), 'Amount' (with the value '0.00' and minus/plus buttons), and 'Description' (an empty text area). At the bottom of the form is an 'Add' button. The top right corner of the application shows a navigation bar with links for 'Share', 'Star', 'Edit', 'Deploy', and a settings menu.

Features of the Project

View Expense

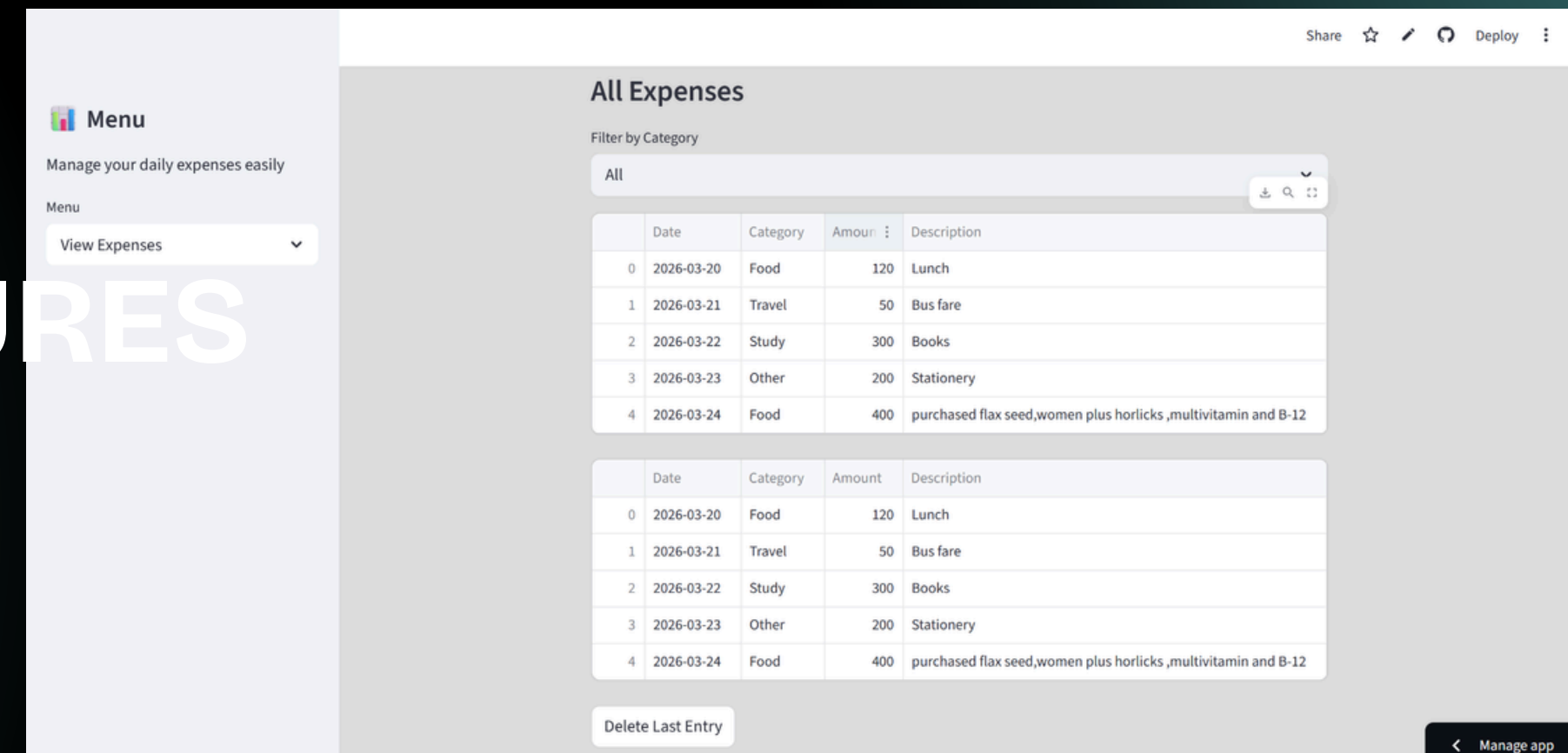
Users can view all recorded expenses in a tabular format. The application also provides filtering options based on categories such as Food, Travel, Stationery, and Others.

Additional functionalities include:

- Deleting the last entry
- Clearing all data

This feature helps users quickly review and manage their spending history.

Interface

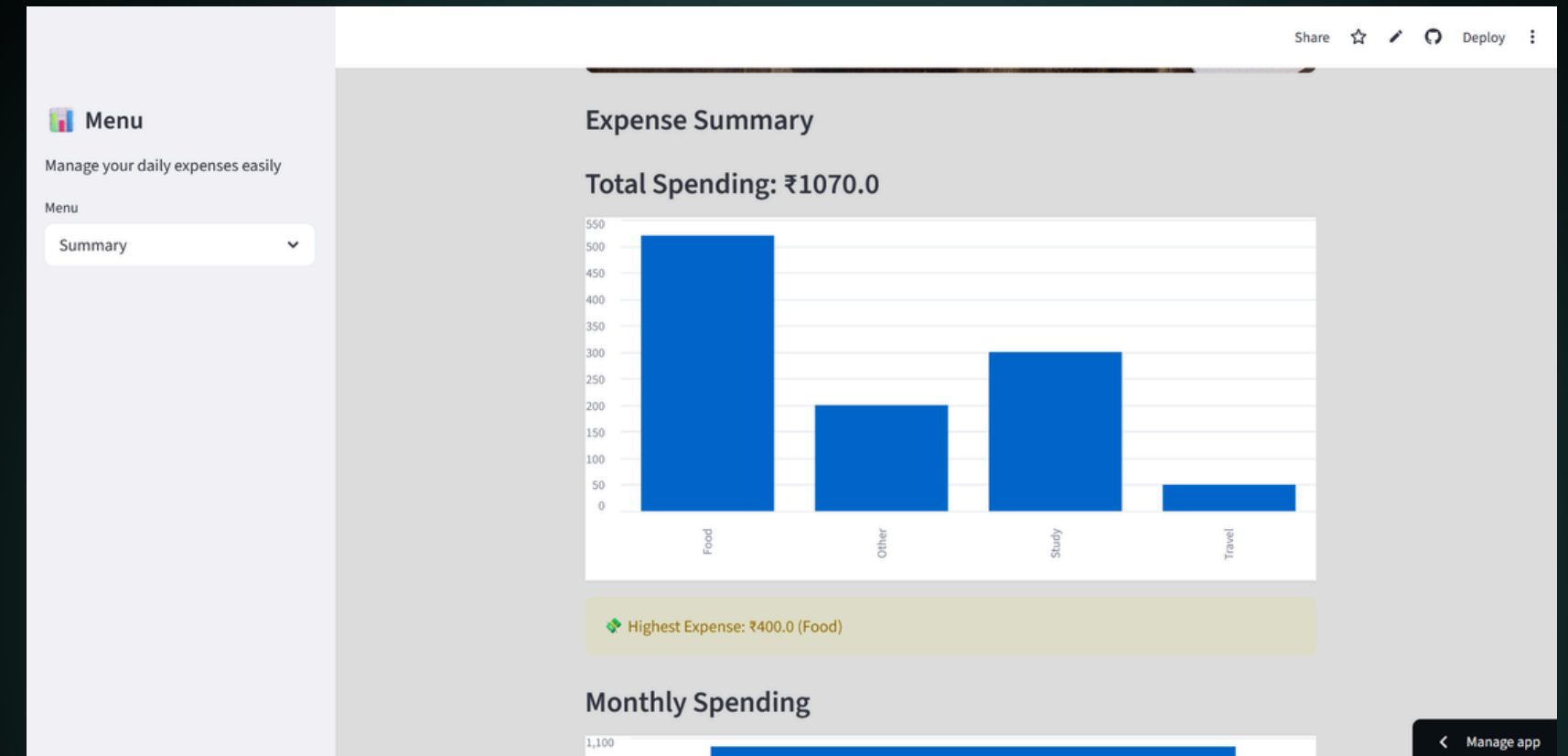


Features of the Project

Summary Dashboard

The application allows users to easily record their daily expenses by entering details such as date, category, amount, and description. This feature ensures proper tracking of all financial activities in a structured manner. It helps users maintain a habit of recording expenses regularly, which is the first step toward effective financial management.

Interface

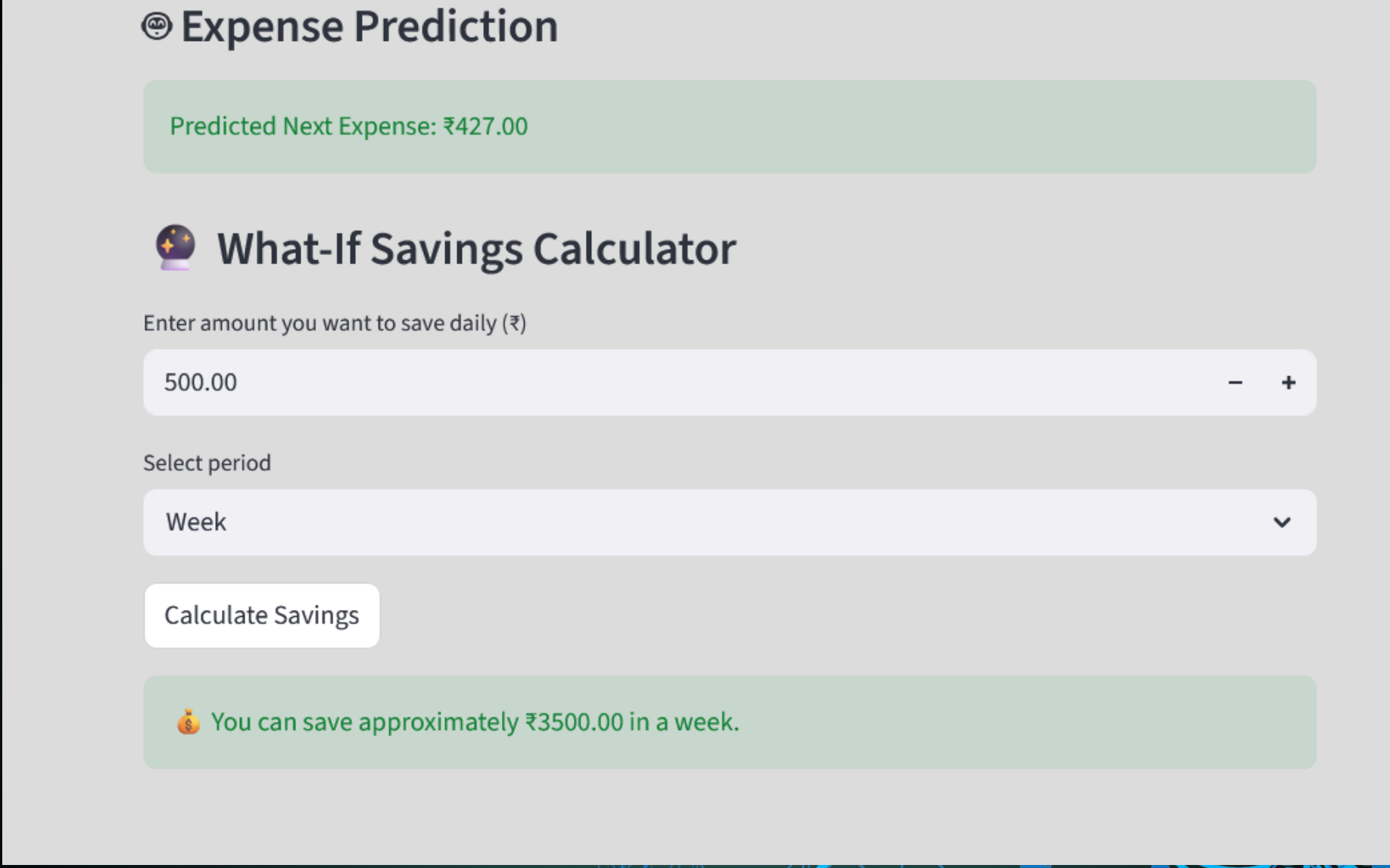


Features of the Project

Expense Predictor + Saving Calculator

- User inputs daily saving amount
- Calculates savings for:
- Weekly
- Monthly
- Yearly periods

Interface



The screenshot displays a web interface for financial calculations. At the top, a header reads "Expense Prediction" with a coin icon. Below it, a green box shows "Predicted Next Expense: ₹427.00". The main section is titled "What-If Savings Calculator" with a magic wand icon. It prompts the user to "Enter amount you want to save daily (₹)" with a text input field containing "500.00" and minus/plus controls. Below this is a "Select period" dropdown menu currently set to "Week". A "Calculate Savings" button is positioned below the dropdown. At the bottom, a green box displays the result: "You can save approximately ₹3500.00 in a week." accompanied by a money bag icon.

Expense Prediction

Predicted Next Expense: ₹427.00

What-If Savings Calculator

Enter amount you want to save daily (₹)

500.00 - +

Select period

Week

Calculate Savings

You can save approximately ₹3500.00 in a week.

Features of the Project

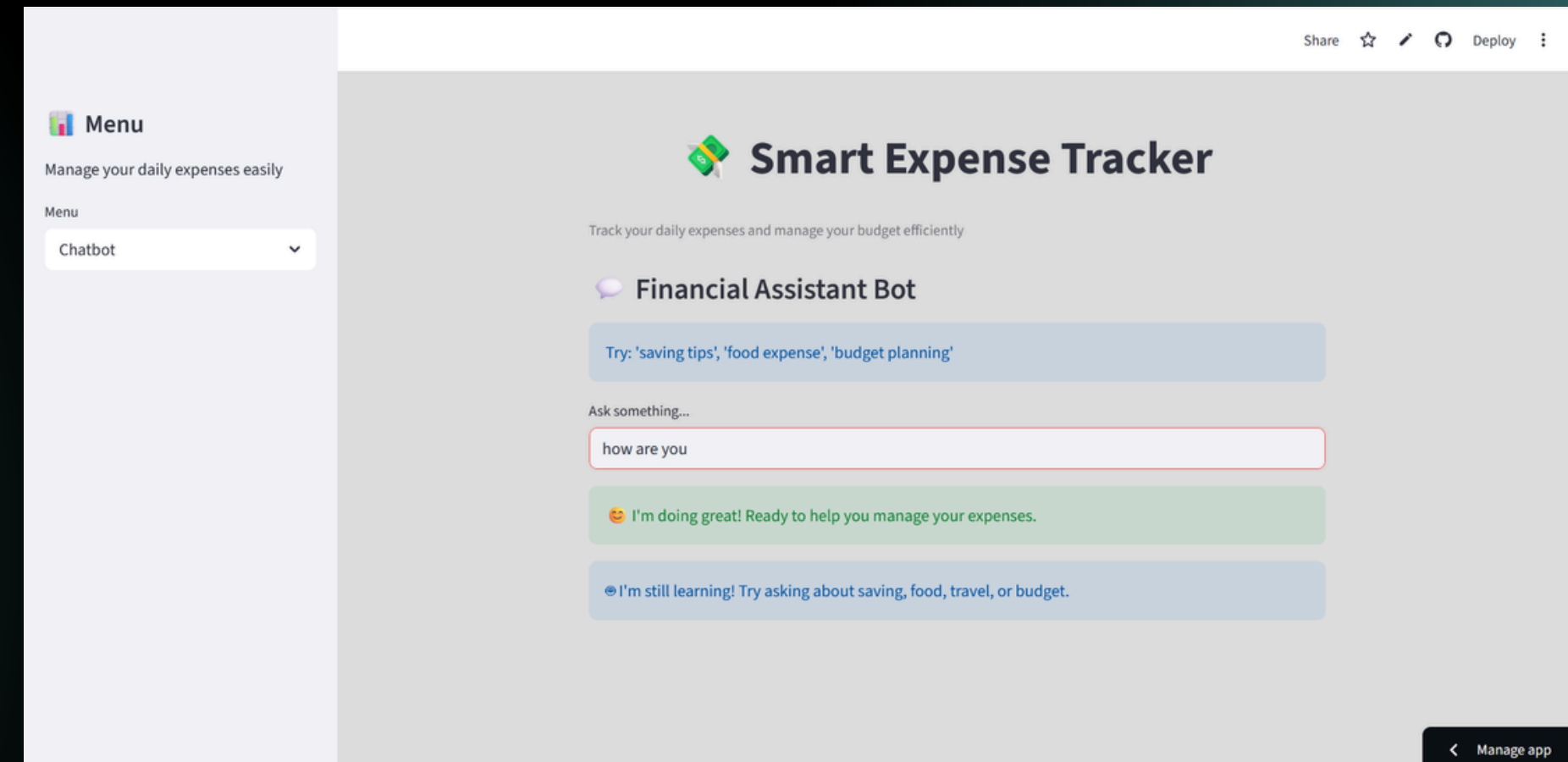
Chatbot

The chatbot can:

- Suggest saving tips
- Provide budgeting advice
- Respond to basic financial queries

This makes the application more interactive and user-friendly.

Interface



Technologies Used

- Python Core programming
- Streamlit Web app development
- Pandas Data handling
- NumPy Numerical operations
- Scikit-learn Machine Learning model
- CSV File Data storage

Methodology

- User inputs stored in a CSV file (expenses.csv)
- Data Processing
- Data is loaded using Pandas
- Preprocessing includes date conversion and grouping
- Visualization
- Bar charts used for category-wise and monthly analysis
- Machine Learning Implementation
- Linear Regression model is used
- Input: Number of days
- Output: Predicted expense
- Chatbot Logic
- Rule-based keyword matching
- Provides predefined financial advice
- What-if Analysis
- Mathematical calculation based on user input

Learning Outcomes

- Through this project, the following skills were developed:
- Understanding of real-world problem solving using AI/ML
- Experience with Streamlit app development
- Knowledge of Machine Learning implementation
- GitHub version control and commits
- Deployment of applications on cloud platforms
- Debugging and dependency management

- Conclusion

The Smart Expense Tracker successfully addresses the problem of personal finance management by combining data tracking, visualization, and predictive analysis.

The project demonstrates how AI/ML concepts can be applied in real-life scenarios to provide meaningful insights. It is simple, practical, and useful for everyday users, especially students.



• Future Scope

1. Integration with database (MySQL/Firebase)
2. Advanced ML models for better predictions
3. User authentication system
4. Mobile app version
5. Voice-based chatbot

• References

- Streamlit Documentation
 - Scikit-learn Documentation
 - Pandas Documentation
 - NumPy Documentation
- 

Thank You



This project was developed as part of the AI/ML course.

Thank you for reviewing my project.

Name: Gauri Gupta (25BAI11091)

Course: Fundamentals of AI and ML Evaluated Course Project